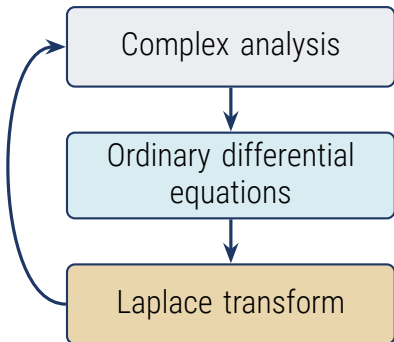




MAT290: Advanced Engineering Mathematics

Lecture 5: Ordinary Differential Equations

Dr. Erfan Meskar

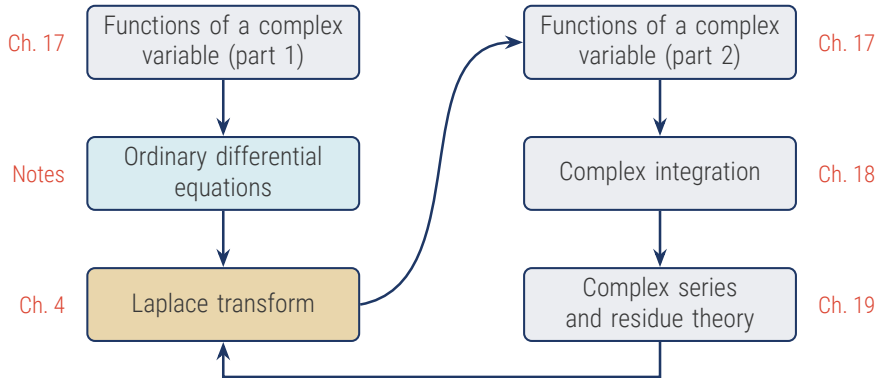
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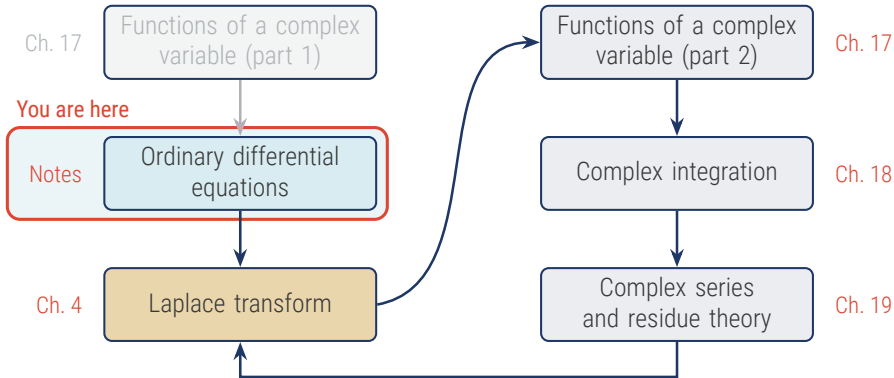


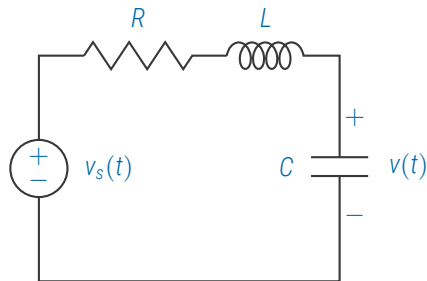
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Our Progression in This Course



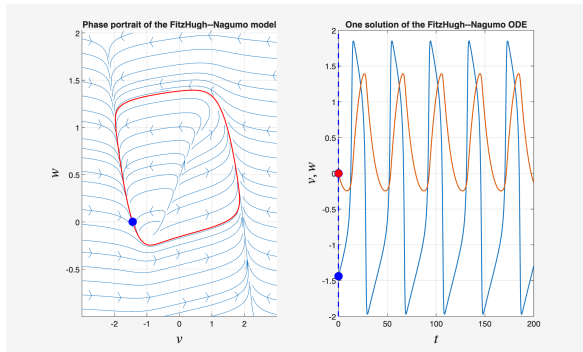
Our Progression in This Course





$$LC\ddot{v} + RC\dot{v} + v = v_s(t).$$

- This is a **second-order, linear, constant-coefficient** ODE.



$$\dot{v} = v - \frac{v^3}{3} - w + I,$$
$$\dot{w} = \varepsilon(v + a - bw).$$

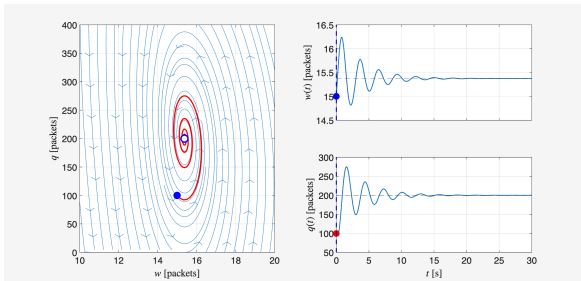
- This is a **nonlinear state-space model** with states (v, w) and output v .



$$\text{Armature: } L_a \frac{di_a}{dt} + R_a i_a + K_e \dot{\theta} = u(t),$$

$$\text{Rotor: } I \ddot{\theta} + b \dot{\theta} - K_t i_a = 0.$$

- The PMDC model consists of **two coupled ODEs**; the PID controller uses measurements of θ to determine $u(t)$.



$$\dot{w} = \frac{1}{R} - \frac{k}{2R} w^2 q,$$

$$\dot{q} = \frac{N}{R} w - C.$$

- This is a **nonlinear state-space model** with states (w, q) and output q , the router queue length.

A classification of ODEs